Periconceptional Folic Acid and Neural-Tube Defects

Randomized and systematic-review evidence shows periconceptional folic acid substantially reduces neural-tube-defect risk; this does not generalize to all anomalies or to intelligence.

DE: HIGH-NTD; LOW-OTHER-ANOMALIES-COGNITION | STATUS: STABLE | DOMAIN: PREGNANCY, NUTRITION | REPLICATION: VERI

WHAT WE KNOW (Narrowest Supported Conclusion)

Daily periconceptional folic acid reduces neural-tube-defect (NTD) risk relative to no supplementation, but the clinical dose differs sharply by baseline risk. In high-risk recurrence (prior NTD-affected pregnancy), 4 mg/day cut NTDs from 3.5% to 1.0% (72% relative reduction; RR 0.28, 95% CI 0.12–0.71; absolute risk reduction 2.5 percentage points). For general-population first occurrence, prevention rests on 400–800 mcg/day: a multivitamin with 800 mcg folic acid eliminated NTDs in a Hungarian RCT (0/2,104 vs 6/2,052), and 400 mcg/day of folic acid alone cut first-occurrence NTDs by 79% in high-baseline northern China and 41% in low-baseline southern China. The Cochrane synthesis reports RR 0.31 (95% CI 0.17–0.58; Summary of Findings lists 6,708 across 5 trials). This is a structural embryonic closure outcome within the first ~28 days post-conception, not a general intelligence or "optimal development" claim.

EVIDENCE AND CAUSAL ASSESSMENT

- *Recurrence RCT — Wald et al. / MRC Vitamin Study Research Group (1991, Lancet, N = 1,817 randomized; 1,195 completed pregnancies): Established prevention of NTD recurrence* with 4 mg/day folic acid (RR 0.28, 95% CI 0.12–0.71; risk dropped from 3.5% to 1.0%, 2.5 percentage-point absolute reduction).

- *First-occurrence RCT — Czeizel & Dudás (1992, NEJM, N = 4,156 known outcomes of 4,753 confirmed pregnancies): Tested a periconceptional multivitamin containing 800 mcg* folic acid versus trace elements in women without prior NTDs (0/2,104 vs 6/2,052 NTDs, P = 0.029). Because the intervention combined folic acid with other micronutrients, single-nutrient isolation awaited China's community intervention.

- *First-occurrence single-nutrient cohort intervention — Berry et al. (1999, NEJM, China–U.S. Collaborative Project, N = 247,831 pregnancies): Evaluated 400 mcg/day folic acid alone without other vitamins. Periconceptional use lowered NTDs by 79% in northern China (31,960 pregnancies; 4.8 to 1.0 per 1,000; 85% among women with >80% pill compliance) and 41% in southern China (215,871 pregnancies; 1.0 to 0.6 per 1,000; 40%* with >80% compliance).

- Cochrane pooling — De-Regil et al. (2015, CD007950): Summary of Findings reports RR 0.31 (95% CI 0.17–0.58; listed as 6,708 across 5 trials), pooling four recurrence trials (RR 0.34, 95% CI 0.18–0.64, n = 1,846) with first-occurrence data from the Hungarian randomized series (Czeizel 1994, n = 5,358 randomized). Cochrane new_version on this record is a successor review, not a retraction.

EFFECT SIZE AND PERSISTENCE

Large relative-risk reduction for NTD (72% recurrence reduction with 4 mg; 79% north / 41% south first-occurrence reduction with 400 mcg alone). Persistence is binary at the structural level (neural tube closes cleanly by week 4 post-conception or does not). Absolute risk reduction depends strongly on baseline folate status (4.8 to 1.0 per 1,000 in folate-poor northern China vs 1.0 to 0.6 per 1,000 in folate-replete southern China).

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a for NTD prevention (established by factorial randomization in MRC 1991, Hungarian RCTs, and prospective periconceptional registration in Berry 1999). Baseline fortification, maternal folate status, and early timing (prior to gestational day 28 post-conception) explain variation in absolute benefit. Mixing the 4 mg MRC recurrence trial with 400 mcg general-population dosing or confusing multivitamin vs isolated folic acid introduces translation bias.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean 4 mg/day is appropriate for the general low-risk population, that supplementation after a missed period prevents neural-tube opening failures already past the ~day-28 closure window, or that folic acid is a general brain-optimization tool.

WHAT ACTION IS REASONABLY JUSTIFIED

For the general reproductive-age population capable of pregnancy, 400–800 mcg/day beginning at least one month before conception alongside food-fortification policy is strongly justified. Higher therapeutic dosing (4–5 mg/day under clinician guidance) is reserved for defined high-risk clinical indications: prior NTD-affected pregnancy, valproate or carbamazepine antiepileptic therapy, pre-gestational type 1/2 diabetes, or severe malabsorption syndromes.

VERIFIED SOURCES & DOI IDENTIFIERS

- De-Regil et al. (2015), Cochrane Database of Systematic Reviews (DOI: 10.1002/14651858.CD007950.pub3 (https://doi.org/10.1002/14651858.CD007950.pub3))
- MRC Vitamin Study Research Group / Wald et al. (1991), Lancet (DOI: 10.1016/0140-6736(91)90133-A (https://doi.org/10.1016/0140-6736(91)90133-A))
- Czeizel & Dudás (1992), New England Journal of Medicine (DOI: 10.1056/NEJM199212243272602 (https://doi.org/10.1056/NEJM199212243272602))
- Berry et al. (1999), New England Journal of Medicine (DOI: 10.1056/NEJM199911113412001 (https://doi.org/10.1056/NEJM199911113412001))